

## p3

Exported 2/14/2026, 5:13:45 PM

---

# 3

In [10]:

```
import numpy as np

P = np.array([
    [0, 1, 0, 0, 0],
    [0, 0, 1, 0, 0],
    [0.5, 0, 0, 0.5, 0],
    [0, 0, 0, 0, 1],
    [0, 0, 1, 0, 0]
])

pi = np.array([1/6, 1/6, 1/3, 1/6, 1/6])

print("pi P =", pi @ P)
print("pi   =", pi)
```

```
pi P = [0.16666667 0.16666667 0.33333333 0.16666667 0.16666667]
pi   = [0.16666667 0.16666667 0.33333333 0.16666667 0.16666667]
```

Since  $\pi P = \pi$ , we have a stationary distribution.

In [11]:

```

p0 = np.array([0.5, 0, 0, 0.5, 0])

def evolve(p, P, n):
    for _ in range(n):
        p = p @ P
    return p
# p0 = np.array([1,0,0,0,0])
for n in [3*n for n in range(0, 12)]:
    print(f"p_{n} =", evolve(p0, P, n))

```

```

p_0 = [0.5 0.  0.  0.5 0. ]
p_3 = [0.5 0.  0.  0.5 0. ]
p_6 = [0.5 0.  0.  0.5 0. ]
p_9 = [0.5 0.  0.  0.5 0. ]
p_12 = [0.5 0.  0.  0.5 0. ]
p_15 = [0.5 0.  0.  0.5 0. ]
p_18 = [0.5 0.  0.  0.5 0. ]
p_21 = [0.5 0.  0.  0.5 0. ]
p_24 = [0.5 0.  0.  0.5 0. ]
p_27 = [0.5 0.  0.  0.5 0. ]
p_30 = [0.5 0.  0.  0.5 0. ]
p_33 = [0.5 0.  0.  0.5 0. ]

```

Since  $p_0 = p_3 = \dots p_{3n}$ , the Markov chain is periodic with period  $k = 3$ .